Assignment 1: Fine-Tune, Sanitize, Extract

Trustworthy Large Language Models - Semester I 2026-27 | Final Experimental Report

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Part A - PEFT placement, rank and quantization

Qwen2.5-1.5B-Instruct sanitizer; completion-only instruction tuning.

All A1-A6 runs used the supplied **20,000 train / 2,000 validation** examples in file order, seed 42, 2 epochs, LR 2e-4 with cosine schedule, LoRA dropout 0.05, max length 768, fp16 and global effective batch 16. Validation used deterministic generation and the same entity/leak metrics later used in Part B.

Run	Method	Placement	r	alpha	Trainable M	Train %	Peak GB	Time min	Val F1	Leak
A1	LoRA	q,v	8	16	1.09	0.071	7.47	87.4	0.9148	0.0424
A2	LoRA	q,k,v,o	8	16	2.18	0.141	7.49	91.9	0.9311	0.0387
A3	LoRA	all attn + MLP	8	16	9.23	0.595	7.62	121.4	0.9461	0.0368
A4	LoRA	all attn + MLP	32	64	36.93	2.336	8.21	122.8	0.9509	0.0373
A5	LoRA	all attn + MLP	32	16	36.93	2.336	8.21	107.4	0.9487	0.0376
A6	QLoRA	all attn + MLP	32	64	36.93	3.990	7.45	134.7	0.9526	0.0380

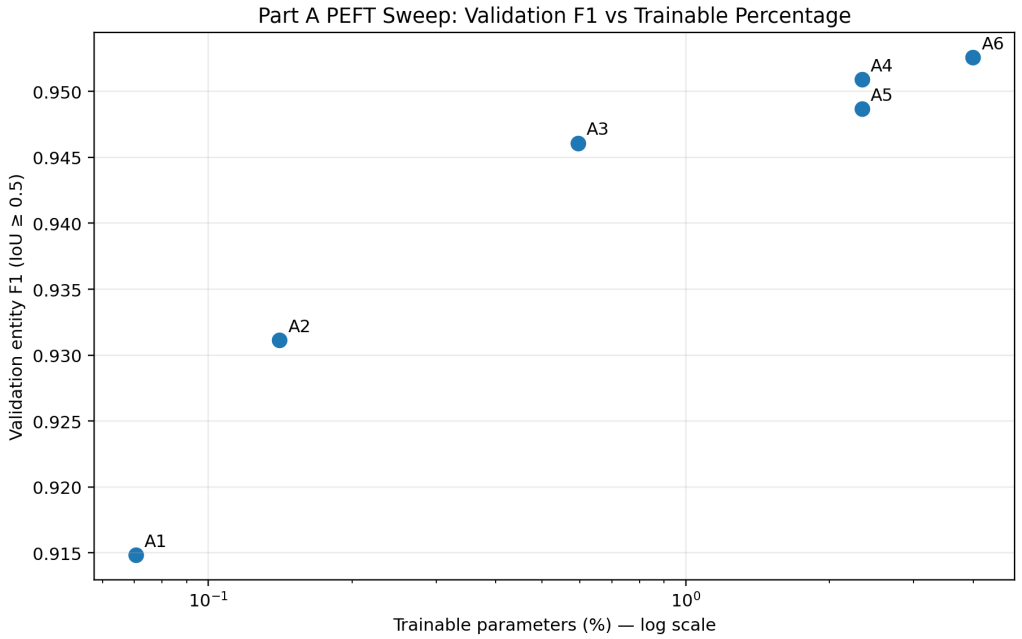


Figure 1. Validation entity F1 versus trainable percentage (log scale). A6 percentage reflects its quantized total-parameter representation.

Selected model: A3 — A3 gives the **lowest validation leak rate (3.675%)** with 94.61% F1 and only 9.23M trainable parameters - about one quarter of the rank-32 alternatives.

Analysis. Placement delivered the largest useful gain: A1->A2 adds 1.63 F1 points with +1.09M trainable parameters, and A2->A3 adds another 1.49 points. Increasing rank A3->A4 quadruples trainable parameters (9.23M->36.93M) for only +0.48 F1 points, showing diminishing returns. At r=32, alpha=64 outperforms alpha=16 by 0.22 F1 points. Matched QLoRA A6 slightly improves F1 over A4 and lowers peak memory by 0.763 GB, but takes longer wall-clock time.

Part B - Sanitizer evaluation on the ID test set

Corrected A3 adapter; 2,000 in-distribution examples.

Span recovery and matching. The model emits masked text. Placeholders are removed to form a literal skeleton; source and skeleton are character-aligned using `difflib.SequenceMatcher(autojunk=False)`. Each placeholder maps to the source interval bounded by the nearest equal literal anchors. Exact and IoU>=0.5 matches are one-to-one; valid same-label IoU pairs are consumed from highest IoU downward. Leak checks are case-insensitive and whitespace-normalized, while over-masking counts changed non-PII source characters.

Table 2. ID sanitizer comparison.

System	Exact P	Exact R	Exact F1	Exact Macro	IoU P	IoU R	IoU F1	IoU Macro	Leak	Over-mask	Well-formed
A3 finetuned	0.927	0.944	0.935	0.920	0.933	0.950	0.941	0.928	0.034	0.0076	0.9995
Regex	0.667	0.211	0.320	0.175	0.713	0.225	0.342	0.212	0.704	0.0037	1.0000
Presidio	0.275	0.189	0.224	0.149	0.288	0.198	0.235	0.155	0.475	0.0389	1.0000

Table 3. A3 per-class performance for the 10 most frequent ID classes.

Class	Support	Exact P	Exact R	Exact F1	IoU P	IoU R	IoU F1
TIME	945	0.949	0.949	0.949	0.954	0.954	0.954
USERNAME	760	0.954	0.950	0.952	0.954	0.950	0.952
SOCIALNUMBER	668	0.949	0.957	0.953	0.957	0.964	0.960
EMAIL	662	0.974	0.976	0.975	0.986	0.988	0.987
IDCARD	651	0.909	0.939	0.924	0.914	0.943	0.928
DRIVERLICENSE	643	0.941	0.939	0.940	0.941	0.939	0.940
LASTNAME1	623	0.857	0.875	0.866	0.865	0.883	0.874
BOD	553	0.957	0.960	0.958	0.969	0.973	0.971
PASSPORT	552	0.901	0.924	0.912	0.908	0.931	0.919
SEX	529	0.954	0.977	0.965	0.959	0.983	0.971

ID result A3 reaches **94.14% IoU-F1** versus 34.25% (Regex) and 23.49% (Presidio), while leak rate falls to **3.44%**. Its 0.76% over-mask rate shows that the privacy gain is not achieved by indiscriminate deletion.

Alignment diagnostics. Across the ID outputs, the evaluator recorded 33 new unknown placeholders, 21 adjacent-group splits, 26 zero-length spans dropped, and only 1 malformed output. Exact F1 (93.50%) is only 0.64 points below IoU-F1, so many residual span errors are localized rather than catastrophic.

Failure modes of span recovery. Large model rewrites, reordered context, repeated/ambiguous anchors, adjacent placeholders without a separating literal anchor, or malformed placeholders can make deterministic alignment ambiguous. These limitations are retained explicitly because alignment quality directly affects entity metrics.

Part B - OOD generalization and error taxonomy

1,000 OOD examples; same metrics and baselines as the ID test.

Table 4. OOD sanitizer comparison.

System	Exact F1	Exact Macro	IoU F1	IoU Macro	Leak	Over-mask	Well-formed
A3 finetuned	0.180	0.132	0.198	0.146	0.242	0.0029	1.0000
Regex	0.104	0.050	0.123	0.059	0.772	0.0007	1.0000
Presidio	0.090	0.039	0.100	0.045	0.327	0.0301	1.0000

Why OOD collapses

56 labels appear in OOD, but **47 are absent from ID**. Those unseen labels account for **2,636/3,251 = 81.1%** of OOD gold entities. A3 IoU-F1 therefore falls from 94.14% to 19.80%, while leak rate rises from 3.44% to 24.18%.

Table 5. A3 ID error taxonomy (diagnostic events are not mutually exclusive).

Error type	Count	Error type	Count
Unmatched predicted entity	872	Examples with over-masking	725
Wrong label	474	Leaked gold entity	440
Missed entity, no overlap	99	Boundary error but IoU>=0.5	83
Boundary error, IoU<0.5	66	Malformed output	1

Error interpretation. Unmatched predictions are the largest failure category, followed by localized over-masking and label confusion. Only 83 matched entities become correct under relaxed IoU, so semantics and missed/unmatched entities matter more than boundary precision alone. The 440 leaked-gold events justify using leak rate, not F1 alone, for model selection.

Baseline behavior. Regex is conservative and therefore has very low over-masking, but that conservatism produces very high leak rates. Presidio masks more aggressively yet still misses many dataset-specific labels. OOD should therefore be interpreted as both distribution shift and ontology shift.

Part C checkpoint

Corrected A3 sanitized **8,007** rows in order with metadata preserved. **7,251** texts changed, 756 remained unchanged, and 13 generations (0.162%) hit the 260-token cap.

Part C - Victim utility and extraction attacks

V-base = untouched Qwen2.5-0.5B; V-raw and V-san use identical victim LoRA settings.

V-raw and V-san use target modules q/k/v/o + gate/up/down, LoRA r=32, alpha=64, dropout 0.05, 4 epochs, LR 2e-4, seed 42 and effective batch 16; only the training text differs. The sanitizer preserved metadata and repeated-input consistency. Critically, **0/20 canaries and 0/425 canary rows retained the planted secret**.

Table 6. Base / V-raw / V-san comparison. Lower P@100 is better for the membership ranking scores.

Metric	V-base	V-raw	V-san
Held-out PPL (64,958 tokens)	14.175	8.778	14.611
Discoverable exact match	0.00%	3.61%	11.76%
Discoverable mean edit similarity	0.257	0.446	0.586
Untargeted P@100: PPL	0%	7%	10%
Untargeted P@100: ref ratio	0%	4%	2%
Untargeted P@100: lowercase ratio	0%	0%	0%
Untargeted P@100: zlib ratio	0%	7%	10%
Untargeted PII recovery	1.83%	6.89%	2.01%
Mean canary exposure (bits)	2.02	13.06	1.33
Targeted member: greedy	1.0%	40.0%	0.5%
Targeted member: any-10	2.0%	47.0%	3.0%
Targeted control: greedy	3.5%	3.5%	3.0%
Targeted control: any-10	10.0%	9.5%	5.5%

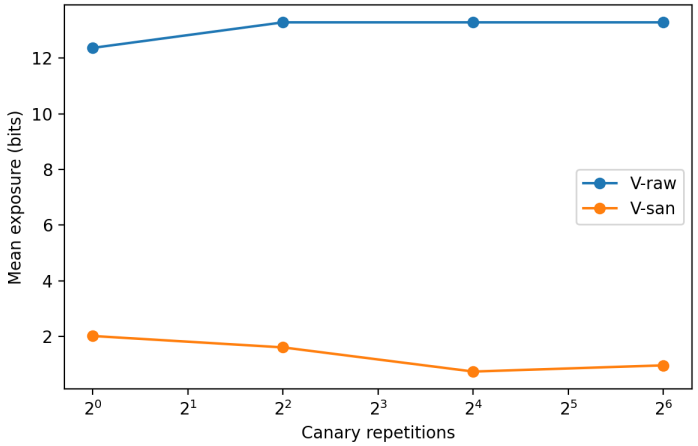


Figure 2. Mean canary exposure versus repetition count. The 9,999-alternative sampled-rank estimator has a maximum measurable exposure of about 13.29 bits.

Attack reading. V-base has zero discoverable exact memorization and zero membership P@100, providing a useful control. V-raw memorizes repeated content aggressively and canary exposure saturates the estimator from 4 repetitions onward. Sanitization sharply reduces privacy-specific leakage, although it does not reduce every generic membership signal: V-san PPL/zlib P@100 are 10%. Sequence memorization and PII leakage are therefore related but not equivalent.

PII-index note. Because `partC_pii_index.jsonl` was not supplied, the recovery index was constructed from Regex + Presidio on one copy of each distinct raw document: 5,020 docs, 4,365 with detected PII and 38,208 detector entries. Recovery rates inherit detector false positives/false negatives.

Part C.3 - Privacy/utility trade-off, traced failures and submission disclosure

Final interpretation; all PII values remain redacted.

Privacy gain Relative to V-raw, sanitization reduces untargeted PII recovery by **70.8%**, mean canary exposure by **89.8%**, greedy member leakage by **98.75%**, and sampled-any10 member leakage by **93.6%**.

Utility cost Held-out PPL rises from **8.78** (V-raw) to **14.61** (V-san), a 66.4% increase. V-san is only 3.1% worse than untouched V-base, so sanitization removes most of the utility gained from raw fine-tuning rather than catastrophically degrading the base model.

Counterintuitive discoverable result. V-san exact-match memorization (11.76%) exceeds V-raw (3.61%). Sanitization converts diverse sensitive spans into more standardized masking patterns, which can make portions of the sanitized corpus easier to reproduce exactly. This does not imply worse privacy because PII recovery, targeted leakage and canary exposure all decrease sharply.

Table 7. Two traced sanitizer failures that directly permitted targeted leakage. Values are redacted.

Trace ID	Doc ID	Label	Survived sanitizer	Greedy hit	Any-10 hit
member 046	doc 00051	SEX	yes	yes	yes
member 055	doc 00062	COUNTRY	yes	no	yes

A third confirmed case (**member_092, TITLE**) also survived sanitization and leaked under sampled-any10. Among 200 member targets, only **3 gold values survived sanitization (1.5%)**, and all three were subsequently extracted by at least one targeted decoding mode. The sole V-san greedy member leak is directly attributable to a sanitizer miss; 3 of the 6 sampled member leaks are likewise traceable to surviving values.

What I would build next. A risk-adaptive two-pass sanitizer: (1) hard-negative fine-tuning focused on observed weak labels such as SEX, COUNTRY, TITLE and OOD-only types; followed by (2) an independent model + Presidio/Regex recheck that remasks suspicious residual spans or abstains for review. Model selection should weight leak rate alongside F1.

Limitations. The Part-C PII index is detector-derived and therefore inherits Regex/Presidio errors. Canary exposure is capped near 13.29 bits by the 10,000-candidate sampled-rank estimator. All attacks target only assignment models/data. Victim/base PPLs were recomputed over the same 64,958 held-out prediction tokens.

TA-accessible adapter folder:
https://drive.google.com/drive/folders/16weDn932DanHp4oOCFelG0h9bcHRDTx4?usp=drive_link

Honour code and disclosure

"Even though I have taken help from the following students and LLMs in terms of discussing ideas and coding practices, all my code is written by me."

Students consulted/discussed: None.

LLM/tool disclosure: ChatGPT (OpenAI) was used for boilerplate/debugging and formatting/organization support; Google Colab and Kaggle notebooks were used for model training/evaluation. Experimental values in this report come from the submitted run artifacts.